Task result overview

Task information

Software version: 1.2.6

Generated UTC: 2026-10-10T15:00:23Z

Task name: reference_600km_24h_report

Analysis duration: 24 h

Time step: 60 s

Thermal network: reference_11_node_gnss_receiver

Outputs from this run

scenario

Main findings

Mean beta angle: 19.46 deg

Eclipse fraction: 36.29 %

RF temperature span: 5.183 K

Semi-synthetic code-bias span: 1.12 m

Orbital-frequency phase delay: 2500.69 s

Maximum energy closure: 1.775e-09 J

Key findings and interpretation

1. Temperature level and variation describe different demands: the maximum is at Zenith face (301.86 K), the minimum at Orbit-left face (211.36 K), and the largest full range at Zenith face (71.95 K).

2. Time-integrated external heat shares are 63.2% direct solar, 3.7% albedo and 33.0% Earth infrared. direct solar contributes most to external absorption; these are heat-input shares, not temperature-response attribution.

3. GNSS RF front end has a full range of 5.556 K, 7.7% of the largest nodal range. Assess response-node stability directly rather than substituting exterior or network-wide extremes.

4. Among links connected to GNSS RF front end, the largest absolute mean net transfer is GNSS RF front end to Spacecraft structure at 6.382 W. Total mean net conduction from the response node is 4.997 W (positive outward). This identifies an active path using heat flow rather than link count.

5. At the orbital fundamental, response temperature lags total external absorption by 41.68 min, or 43.1% of an orbit. Same-time heat-input comparisons omit this phase relationship; the value is not a heat-transport travel time.

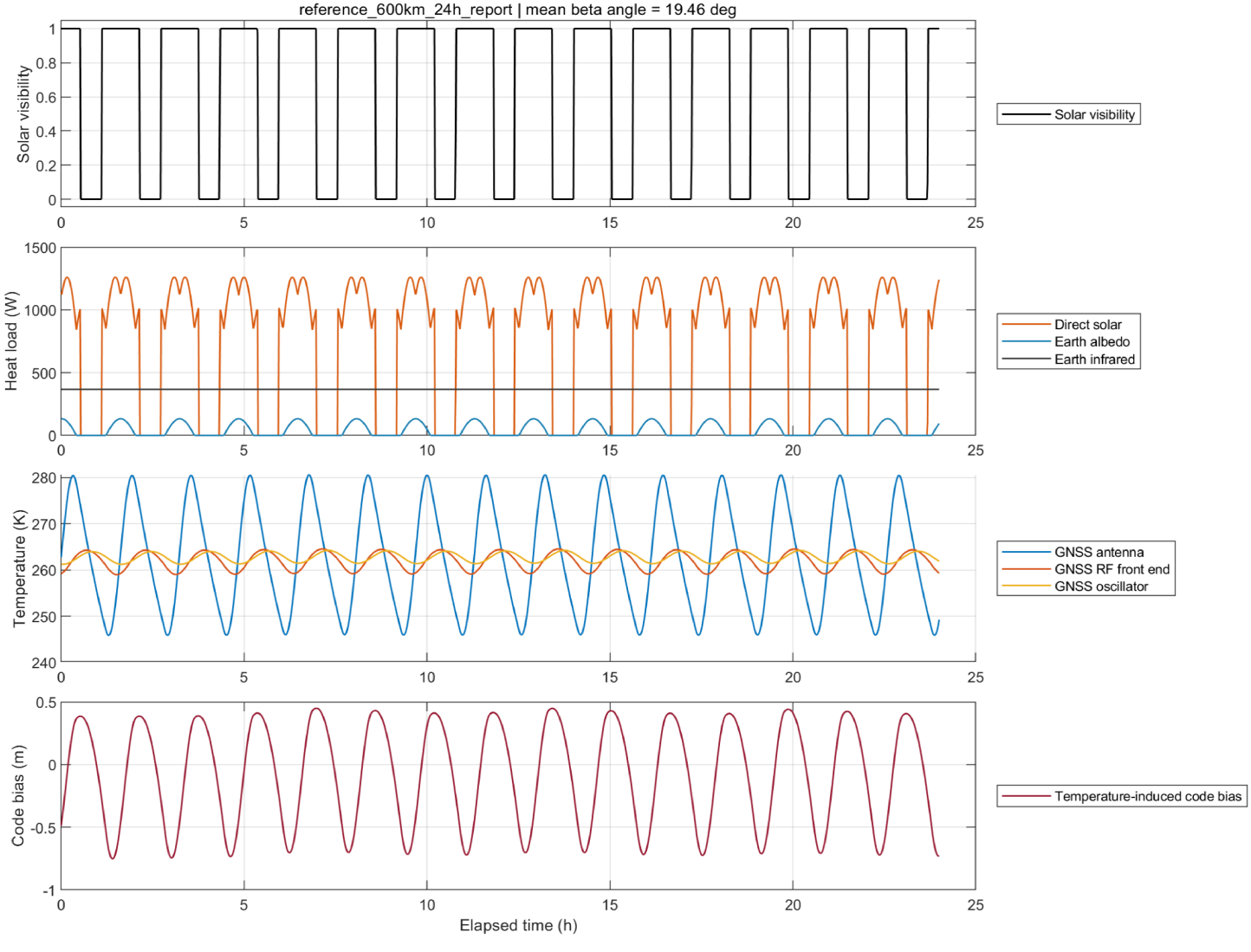
Orbit heating and nodal response

Mean beta angle: 19.46 deg; eclipse fraction: 36.29%.

RF temperature span: 5.183 K; assumed temperature-induced code-bias span: 1.12 m.

Orbital-frequency phase delay: 2500.69 s (41.68 min).

30 illumination transitions occurred over this run.



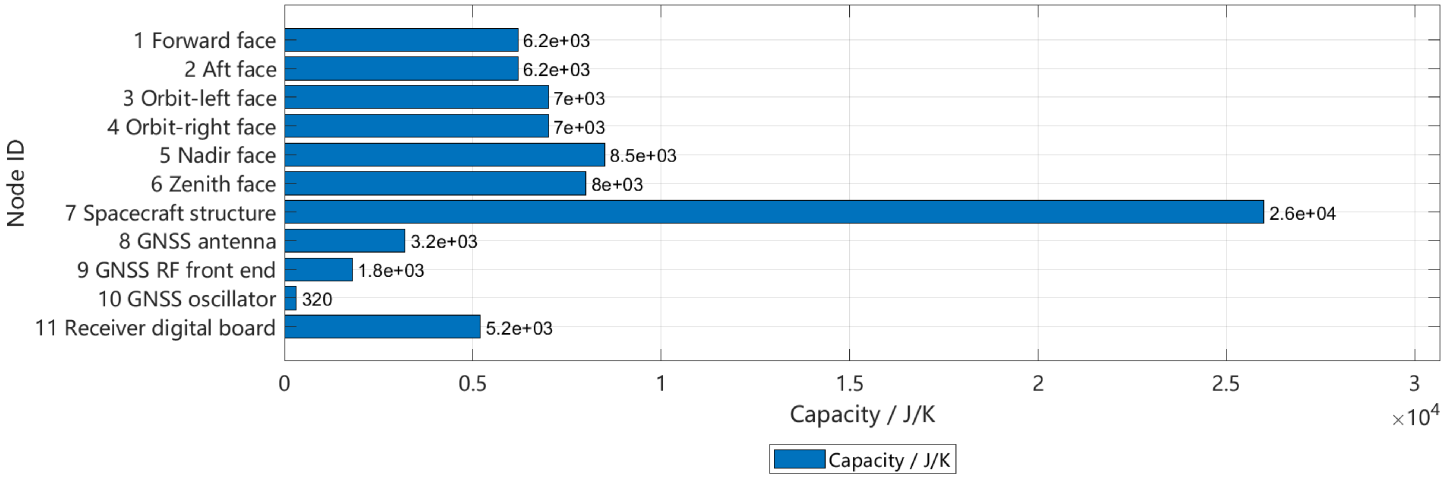
Thermal capacity and internal power

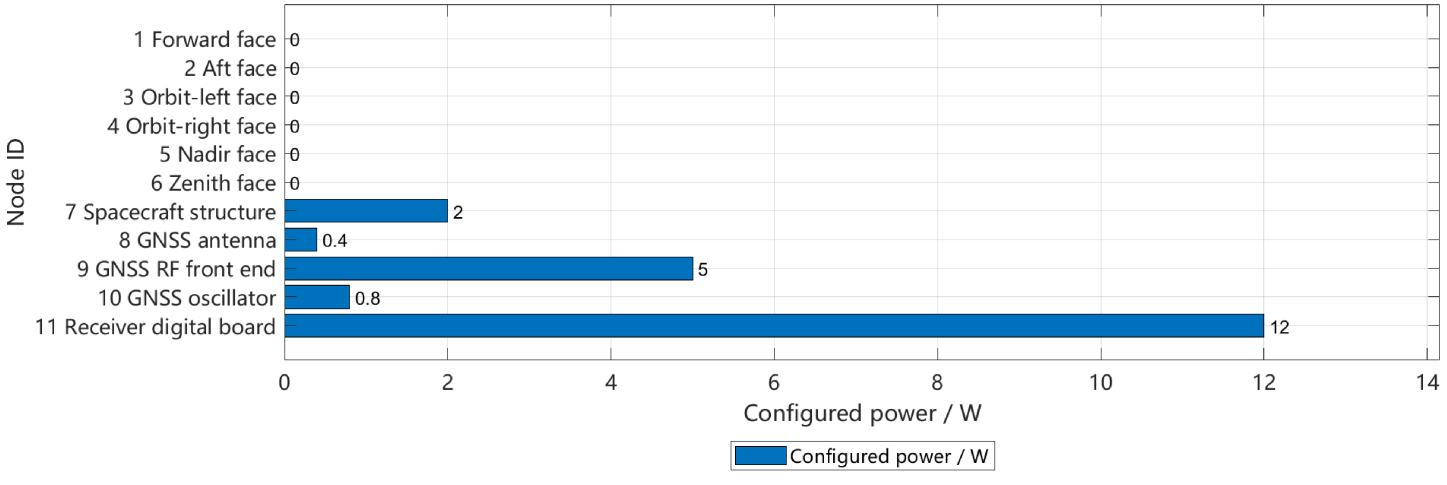
11 nodes; total capacity 79420 J/K; configured internal power 20.2 W.

Spacecraft structure has the largest capacity: 26000 J/K (32.7% of total).

Receiver digital board has the highest configured power: 12 W.

11/11 nodes plotted, selected by full temperature range; all are tabulated.





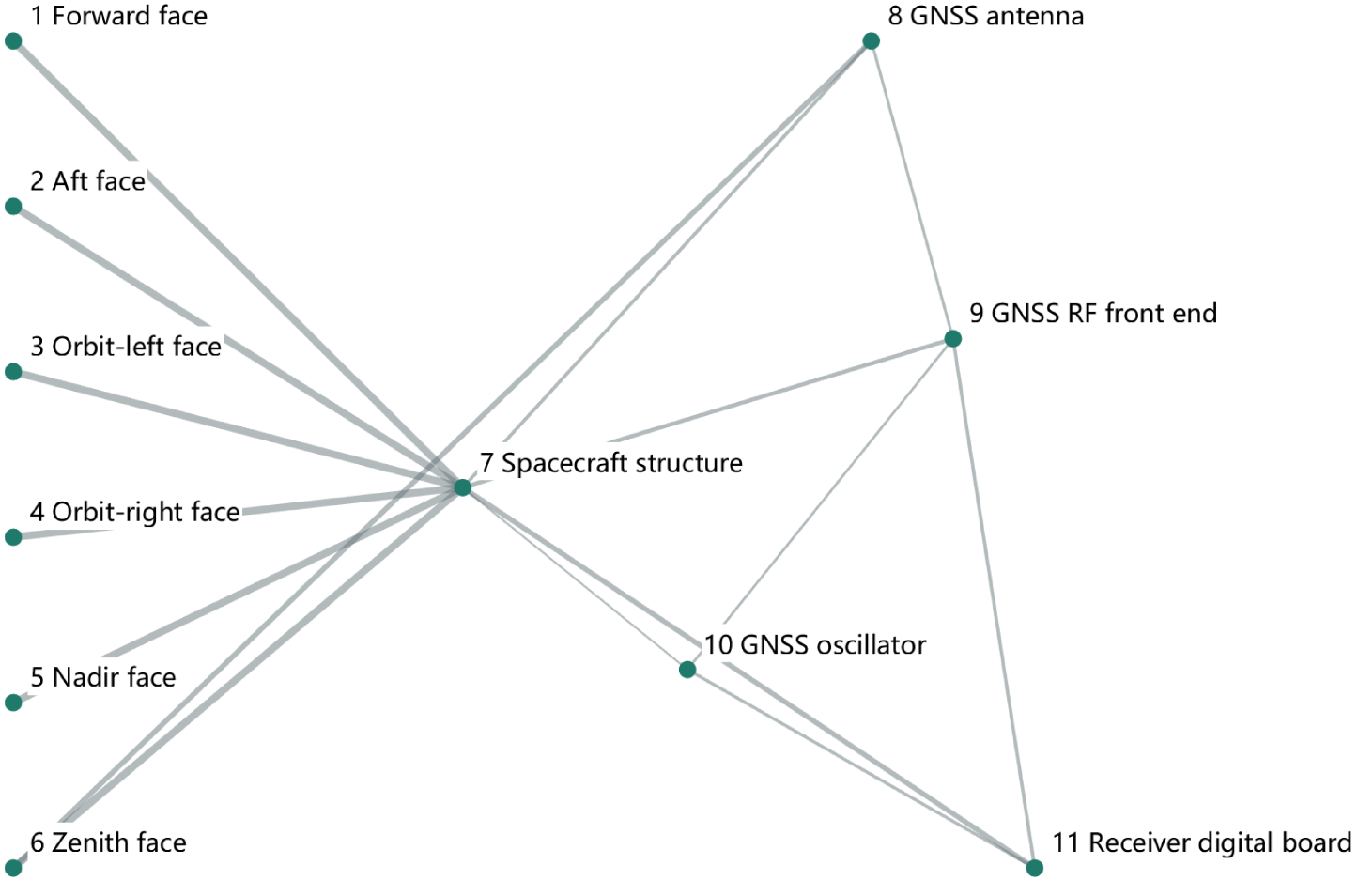
Thermal links and conductance

15 conductive links; Spacecraft structure has the most connections (10).

First 11/11 nodes shown; thicker lines indicate larger conductance.

The layout depicts thermal connectivity, not spacecraft geometry. Conductances are listed in the link CSV.

Among links connected to GNSS RF front end, the largest absolute mean net transfer is GNSS RF front end to Spacecraft structure at 6.382 W. Total mean net conduction from the response node is 4.997 W (positive outward). This identifies an active path using heat flow rather than link count.

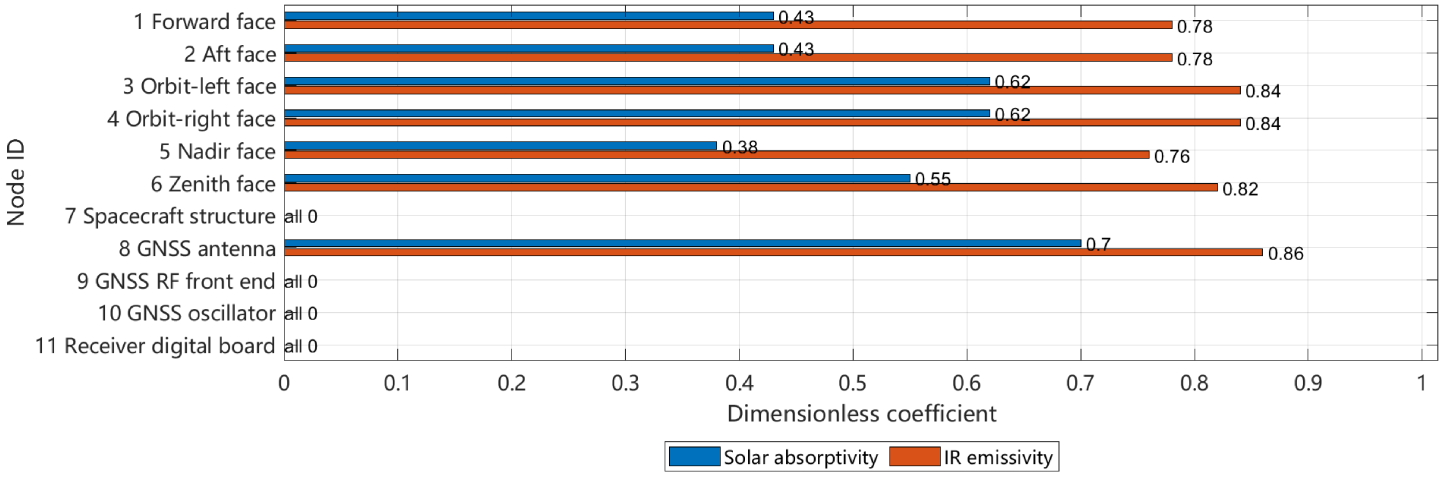


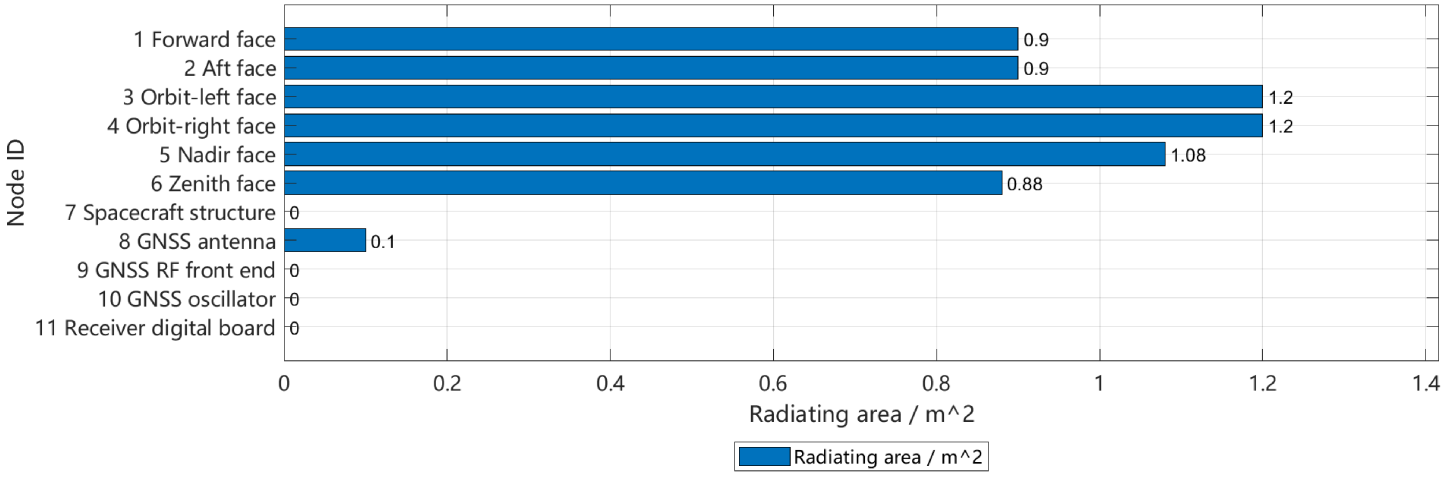
Optical properties and radiating area

Total external radiating area: 6.26 m^2. Zero values are explicitly labelled.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Spacecraft structure, GNSS RF front end, GNSS oscillator, Receiver digital board have zero projected and external radiating areas and exchange heat through conductive links.





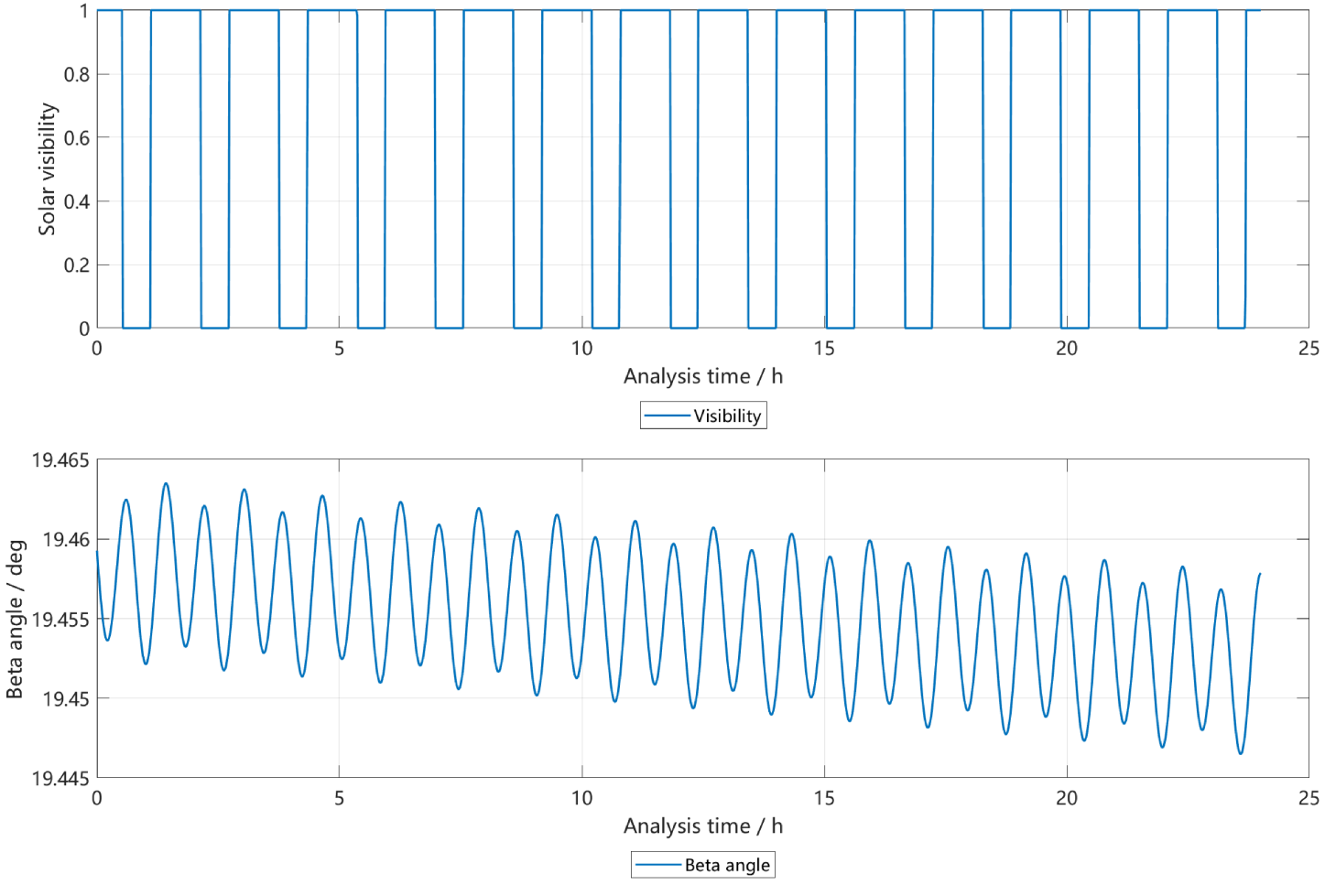
Illumination and eclipse structure

Trapezoidal time weighting estimates partial/full eclipse over 36.32% of the window.

Solar visibility ranges from 0.000 to 1.000; orbital period 96.69 min.

Eclipse uses visibility below 0.999. Time-weighted and epoch-count fractions can differ.

15 eclipse entries and 15 exits; beta angle 19.446 to 19.463 deg.



Environmental heat inputs

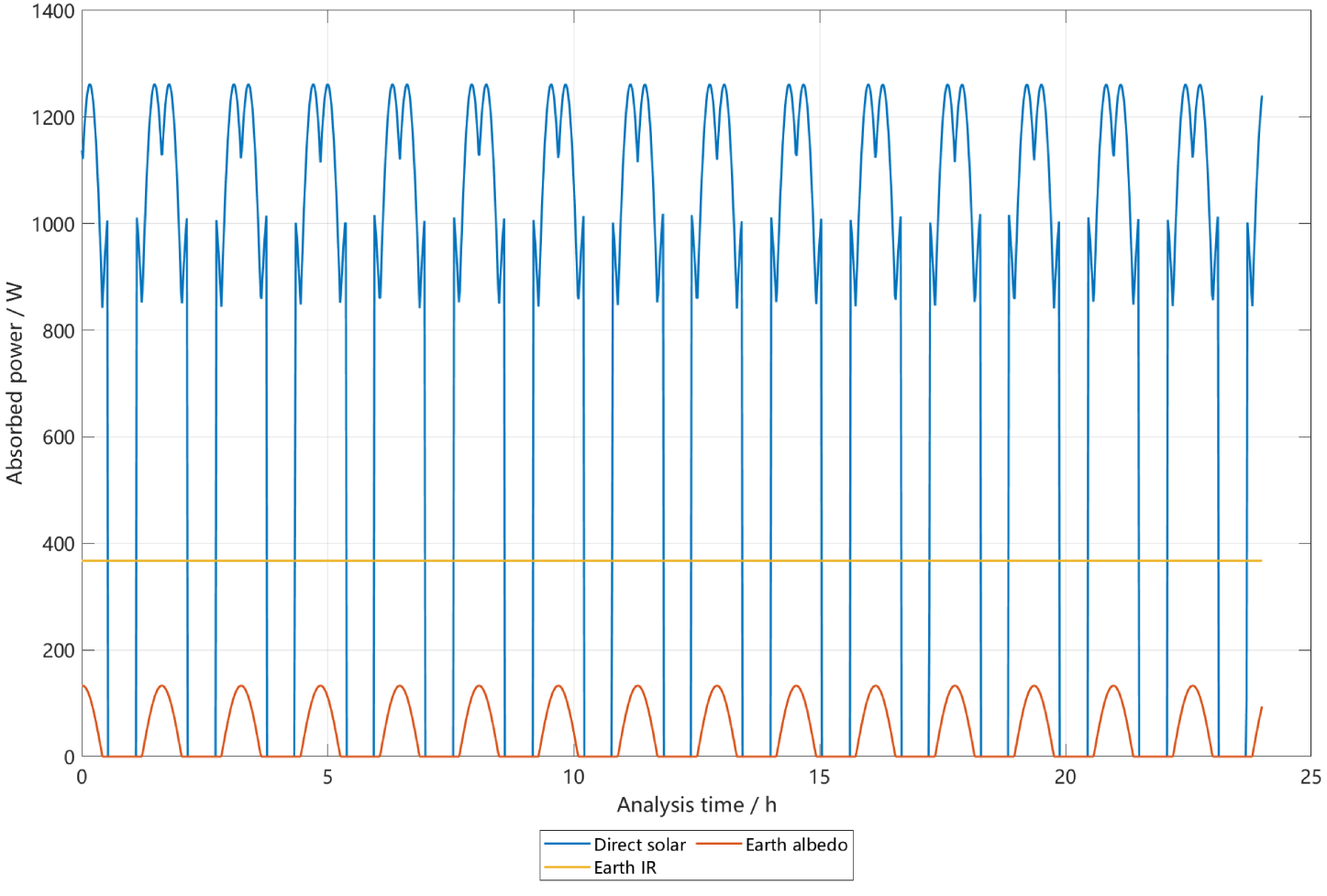
Time-mean absorbed power: solar 704.069 W; albedo 41.743 W; Earth IR 367.422 W.

Curves sum absorbed power over the network, including area and absorption properties.

Solar absorbed power 0.000 to 1261.440 W; Earth IR 367.422 to 367.422 W.

Time-integrated external heat shares are 63.2% direct solar, 3.7% albedo and 33.0% Earth infrared.

direct solar contributes most to external absorption; these are heat-input shares, not temperature-response attribution.



Nodal temperatures and extrema

Window maximum 301.864 K (28.71 deg C, node 6); minimum 211.359 K (-61.79 deg C, node 3).

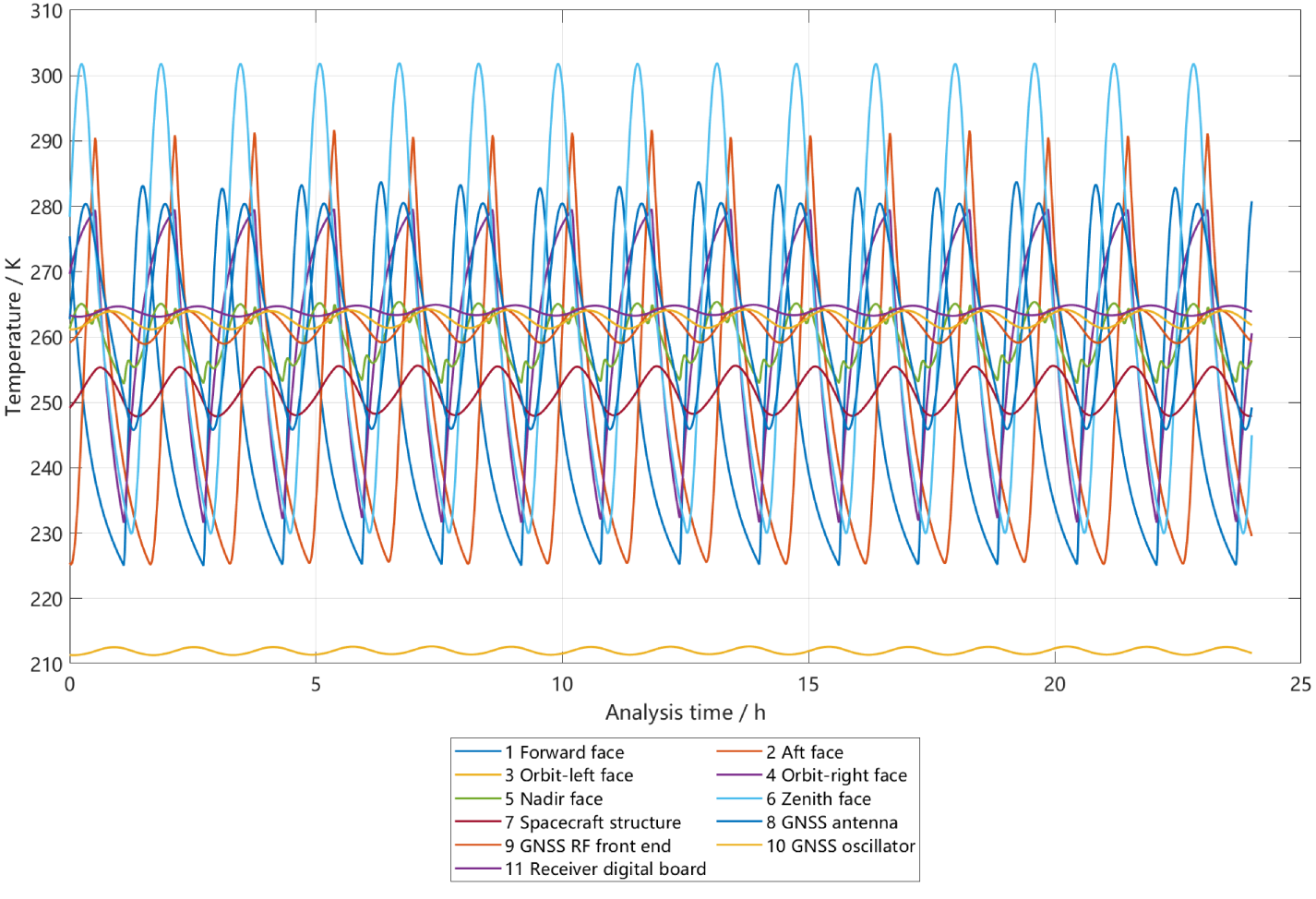
Largest full peak-to-peak range 71.952 K (node 6), distinct from a 95th-5th percentile span.

Hottest node: Zenith face; coldest node: Orbit-left face.

11/11 nodes plotted, selected by full temperature range; all are tabulated.

Temperature level and variation describe different demands: the maximum is at Zenith face (301.86 K), the minimum at Orbit-left face (211.36 K), and the largest full range at Zenith face (71.95 K).

GNSS RF front end has a full range of 5.556 K, 7.7% of the largest nodal range. Assess response-node stability directly rather than substituting exterior or network-wide extremes.

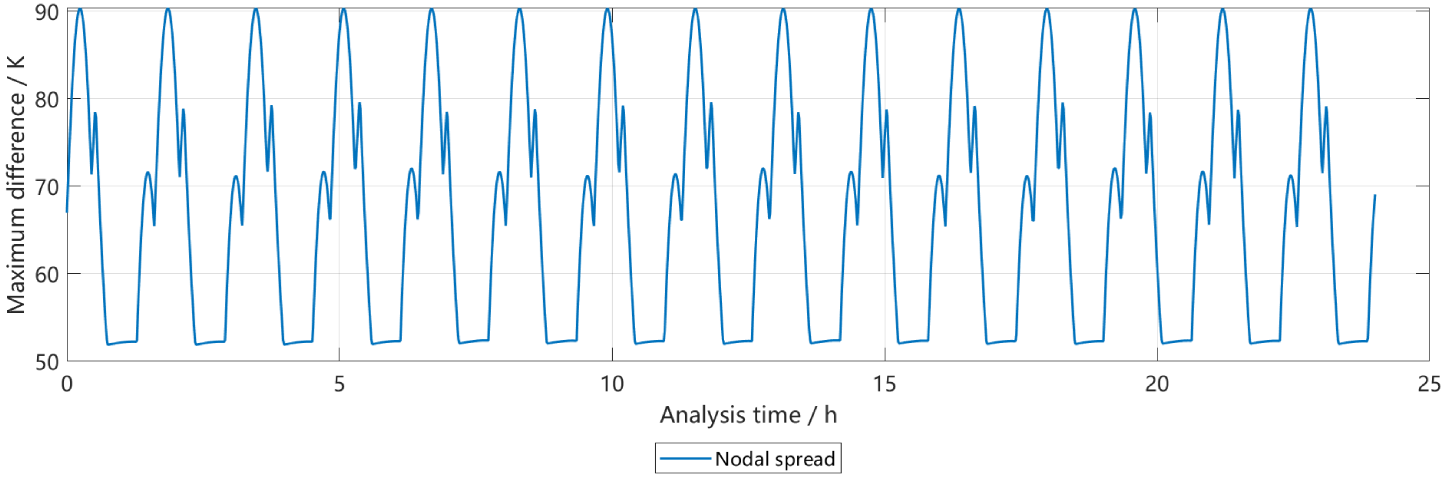


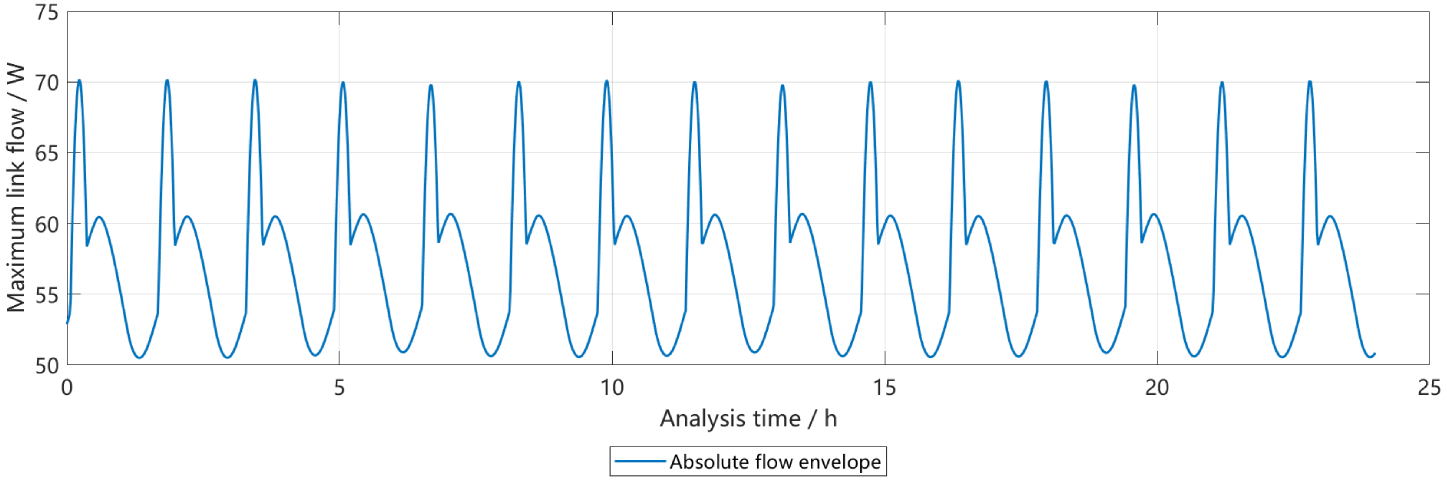
Temperature differences and link heat flow

Maximum simultaneous nodal spread 90.341 K; maximum absolute single-link flow 70.175 W.

Panels show epoch-wise temperature spread and the largest absolute link heat flow.

The envelope takes the largest value across 15 conductive links at each epoch.





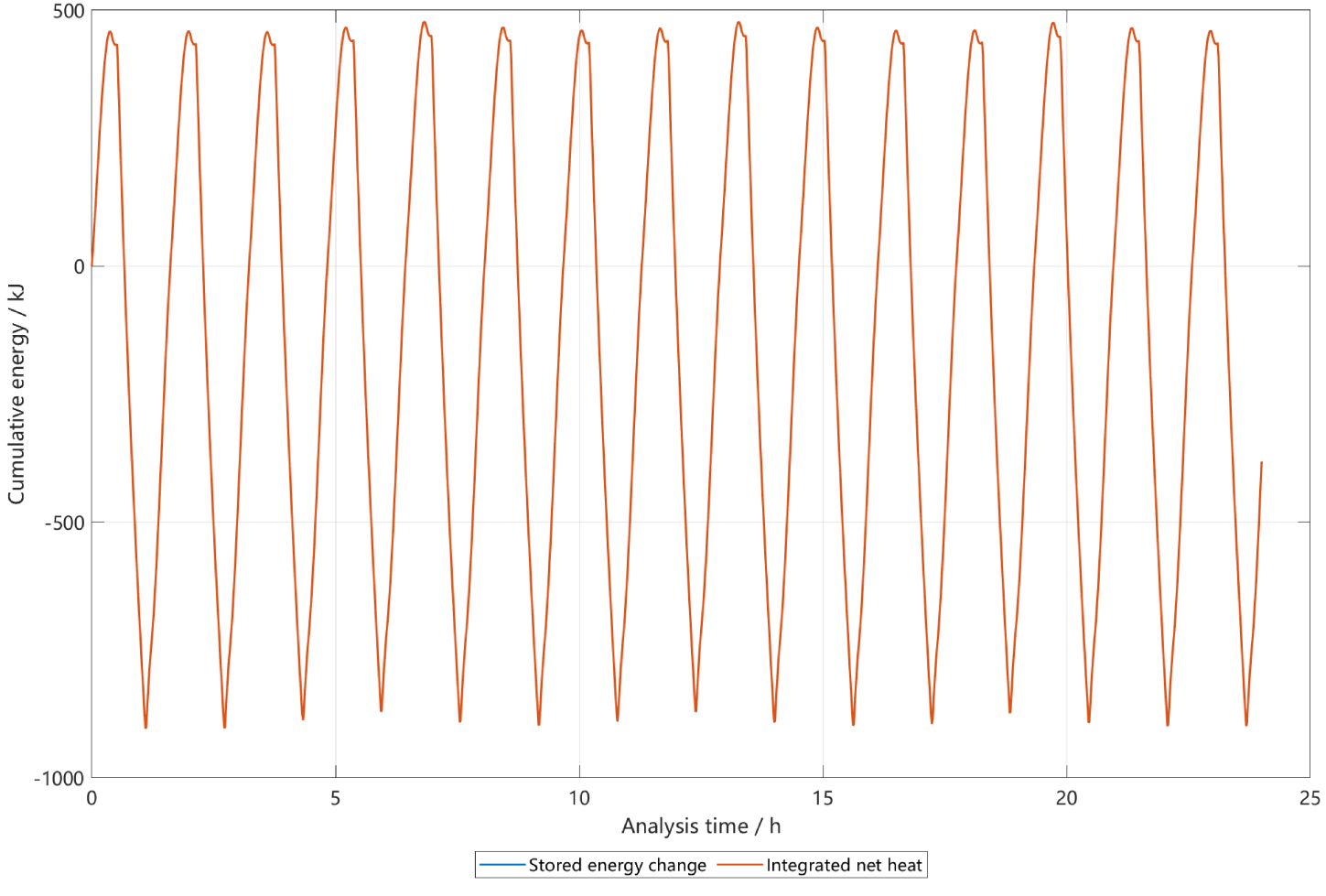
Stored energy and energy balance

After rebasing both at the analysis start, maximum cumulative closure is 4.8938e-08 J.

Stored energy is the sum of capacity times temperature change, relative to analysis start.

Balance: stored energy = external heat + internal heat - radiation + boundary heat.

End-of-window stored-energy change: -381.84 kJ.

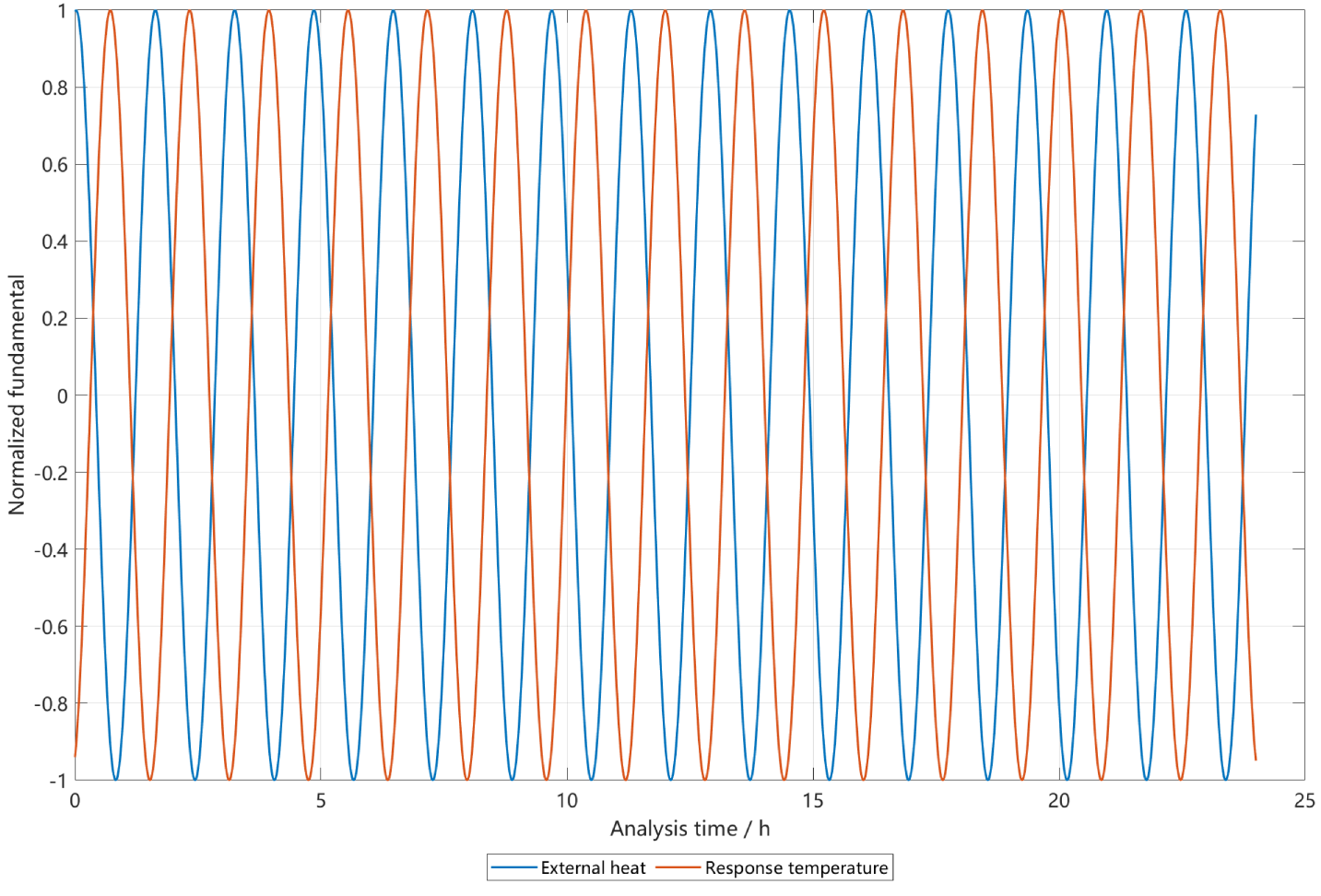


Orbital thermal response

Orbital-frequency phase delay between total external heat and GNSS RF front end temperature: 2500.69 s (41.68 min).

Fundamental amplitudes: 768.916 W and 2.631 K. Separate half-window estimates differ by 2.62 s. Offset, trend and three harmonics are fitted jointly at actual epochs. Curves show amplitude-normalized fundamental components.

At the orbital fundamental, response temperature lags total external absorption by 41.68 min, or 43.1% of an orbit. Same-time heat-input comparisons omit this phase relationship; the value is not a heat-transport travel time.



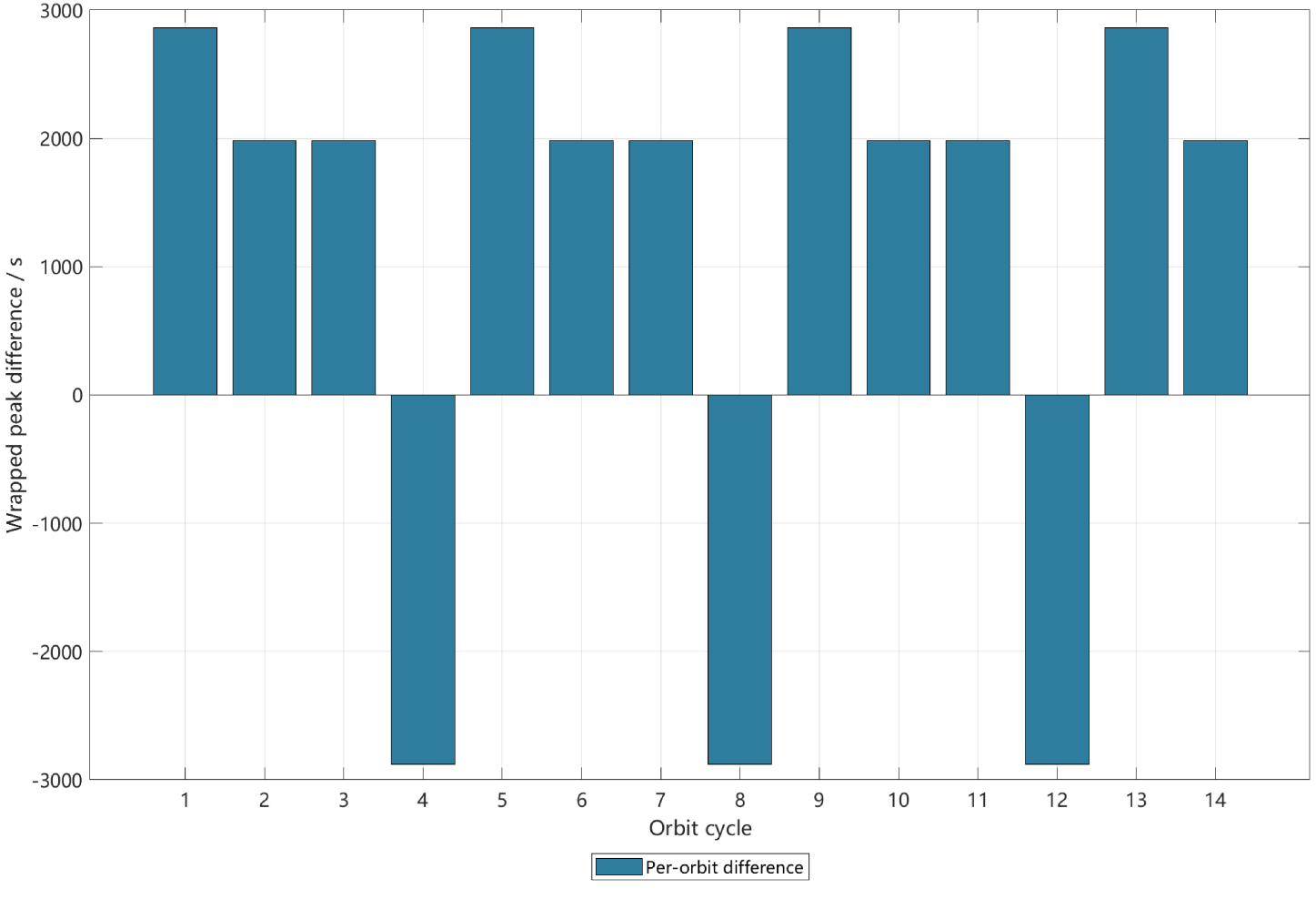
Per-orbit peak timing

Response node: GNSS RF front end.

Temperature peaks follow heat peaks in 11 cycles, precede them in 3, and coincide in 0; wrapped differences range from -2880 to 2861 s and are shown per cycle.

Network external-heat peaks versus response-temperature peaks; differences wrap to +/- half an orbit (2900.6 s).

Positive values indicate later wrapped temperature peaks; negative values indicate earlier peaks. Both are retained.



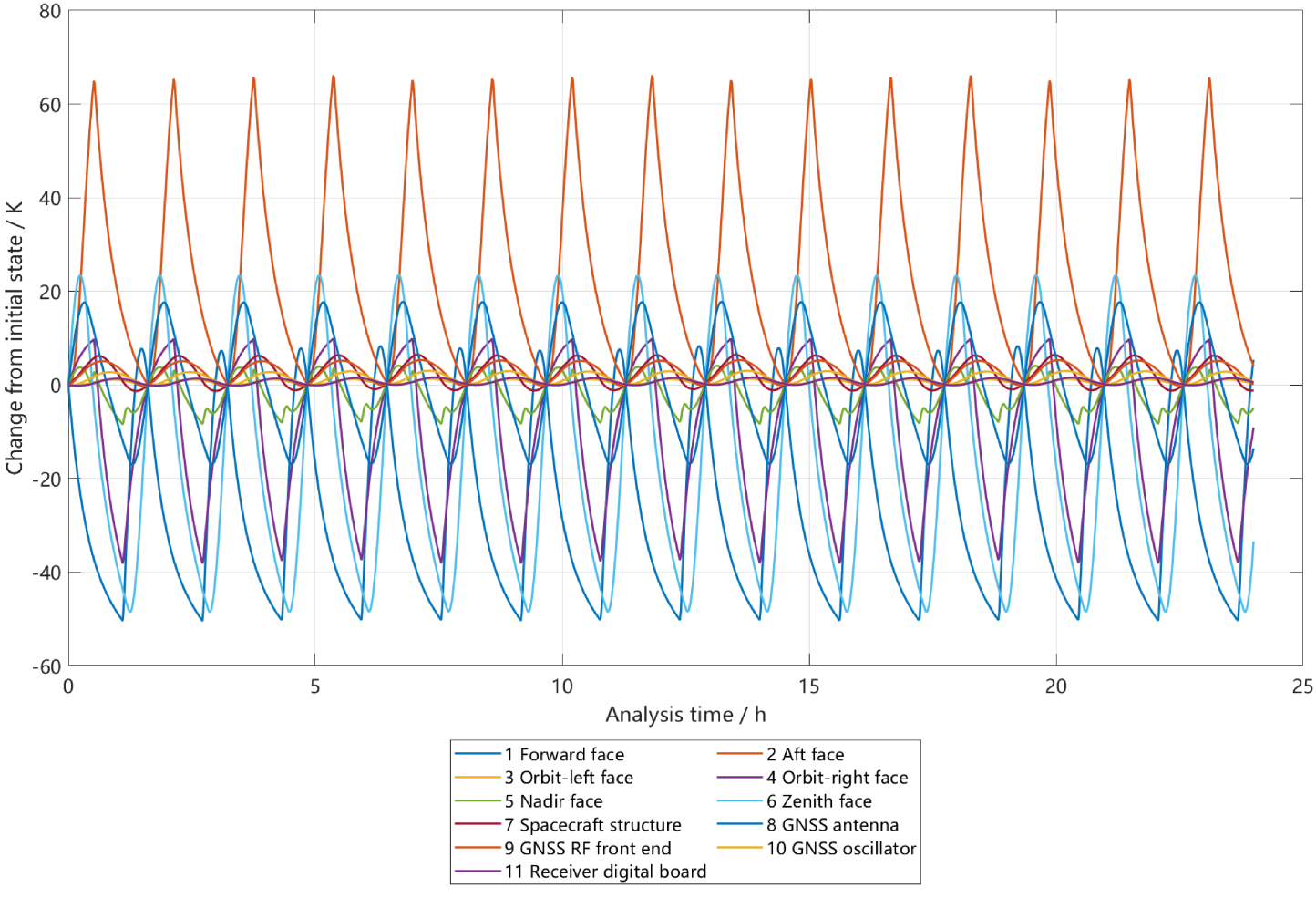
Initial state and temperature drift

Frozen-Sun initialization: 17 cycles; periodic difference 0.00045696 K; met the configured criterion.

Largest absolute end-minus-start temperature change 33.415 K.

Endpoints are 14.893 orbits apart; curves are relative to analysis-start temperatures.

11/11 nodes plotted, selected by full temperature range; all are tabulated.



Time-grid refinement audit

Independent reruns keep the model and scenario fixed and compare exact common epochs.

60/30 s grids: maximum difference 1.186 K; response-node difference 0.2349 K.

Orbital-frequency delays: 2500.69 and 2500.65 s.

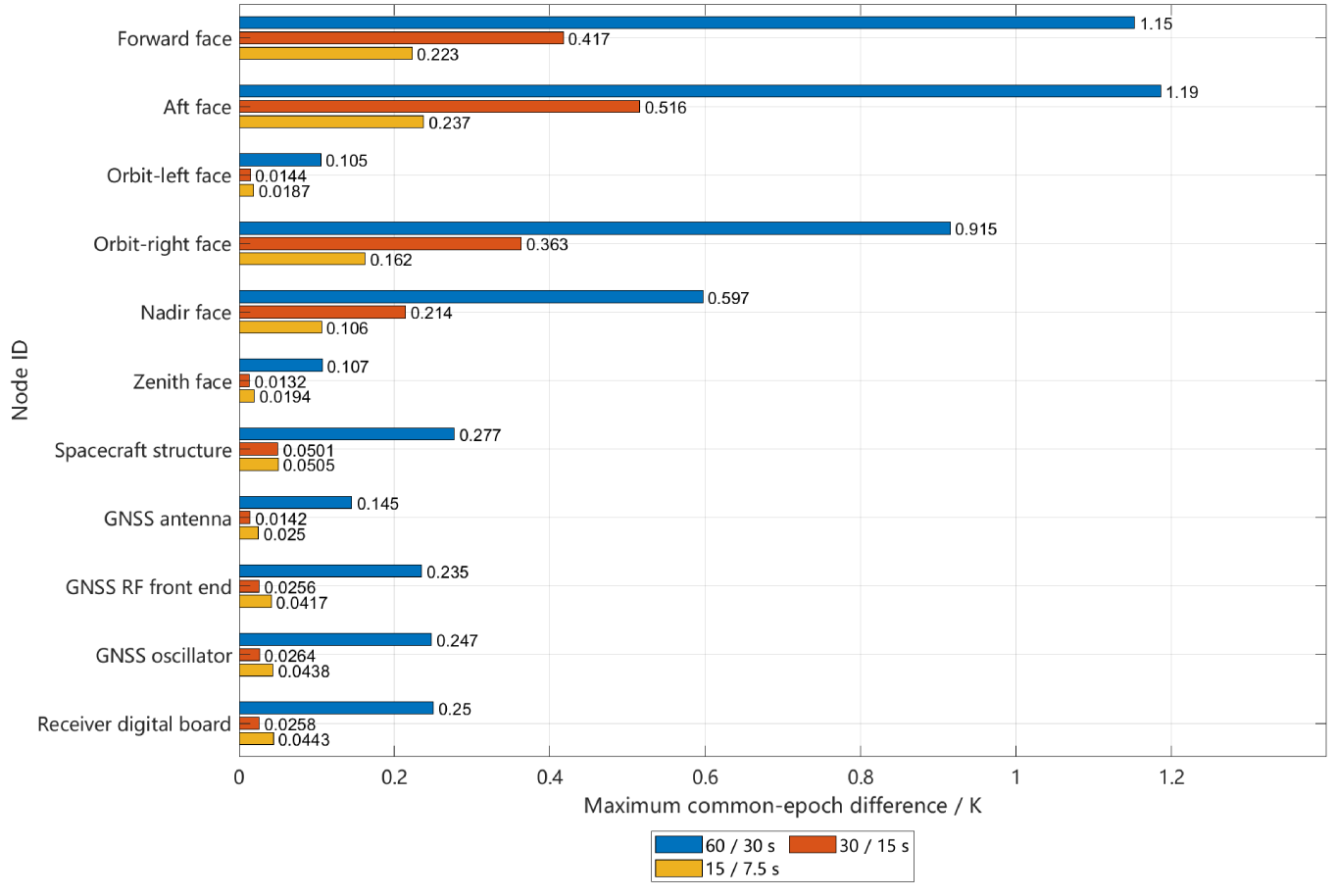
30/15 s grids: maximum difference 0.5155 K; response-node difference 0.0256 K.

Orbital-frequency delays: 2500.65 and 2500.70 s.

15/7.5 s grids: maximum difference 0.2375 K; response-node difference 0.04169 K.

Orbital-frequency delays: 2500.70 and 2501.57 s.

The finest pair differs by up to 0.2375 K across all nodes and 0.04169 K at the response node; these are measured refinement differences.



All-node statistics

Temperature: K; capacity: J/K; configured power: W. Means are time-weighted.

1: Forward face

Min 225.028 | Mean 248.909 | Max 283.724 | Full range 58.696

Capacity 6200 | Internal power 0 | End minus start 5.360 K

2: Aft face

Min 225.225 | Mean 249.284 | Max 291.666 | Full range 66.441

Capacity 6200 | Internal power 0 | End minus start 4.068 K

3: Orbit-left face

Min 211.359 | Mean 212.005 | Max 212.689 | Full range 1.330

Capacity 7000 | Internal power 0 | End minus start 0.295 K

4: Orbit-right face

Min 231.574 | Mean 259.904 | Max 279.564 | Full range 47.990

Capacity 7000 | Internal power 0 | End minus start -9.101 K

5: Nadir face

Min 252.850 | Mean 259.558 | Max 265.379 | Full range 12.529

Capacity 8500 | Internal power 0 | End minus start -4.866 K

6: Zenith face

Min 229.912 | Mean 262.137 | Max 301.864 | Full range 71.952

Capacity 8000 | Internal power 0 | End minus start -33.415 K

All-node statistics (continued)

7: Spacecraft structure

Min 247.871 | Mean 251.652 | Max 255.642 | Full range 7.771

Capacity 26000 | Internal power 2 | End minus start -1.160 K

8: GNSS antenna

Min 245.796 | Mean 262.748 | Max 280.520 | Full range 34.724

Capacity 3200 | Internal power 0.4 | End minus start -13.522 K

9: GNSS RF front end

Min 258.937 | Mean 261.946 | Max 264.492 | Full range 5.556

Capacity 1800 | Internal power 5 | End minus start 0.131 K

10: GNSS oscillator

Min 261.160 | Mean 262.727 | Max 264.206 | Full range 3.046

Capacity 320 | Internal power 0.8 | End minus start 0.642 K

11: Receiver digital board

Min 263.121 | Mean 264.036 | Max 264.940 | Full range 1.819

Capacity 5200 | Internal power 12 | End minus start 0.582 K

Inputs and result details

Scenario name: reference_600km_24h_report

Duration: 86400 s

Time step: 60 s

Fixed warmup setting (non-convergence mode): 30 orbit

Node count: 11 node

Conductance range: [0, 1.4] W/K

Internal power range: [0, 12] W

Start epoch UTC: 2024-03-20 12:00:00

Altitude: 600000 m

Inclination: 97.6 deg

Attitude mode: nadir

Trajectory coordinate system: Geocentric equatorial axes (built-in approximate orbit model)

Telemetry comparison: Not run (no telemetry data imported for this task)

Computation checks and audit record

Numerical and configuration checks

Finite scenario temperatures: Pass

Telemetry comparison: Not run (no telemetry data imported for this task)

Assessment scope

This report evaluates the configured thermal simulation and numerical checks. Device temperature-limit compliance is outside this assessment.

Audit files

MAT results, metric CSV, Markdown, manifest, and figures accompany this PDF.

Software version: 1.2.6

Task conclusions

1. Prioritize the high-temperature level at Zenith face, low-temperature level at Orbit-left face, and variation at Zenith face in design review: extrema 301.86/211.36 K and largest range 71.95 K.

2. Prioritize direct solar in environmental-input review: it supplies 63.2% of absorbed external heat. Interpret temperature response together with the heat-transfer paths and orbital phase.

3. The response component GNSS RF front end spans 258.937 to 264.492 K, with a full range of 5.556 K. Use these component values separately from spacecraft extrema.

4. Prioritize the GNSS RF front end / Spacecraft structure contact in conductance review: mean net transfer is 6.382 W. This ranking uses observed model heat flow, not a conductance-perturbation experiment.

5. At the orbital fundamental, response temperature lags total external absorption by 41.68 min, or 43.1% of an orbit. Same-time heat-input comparisons omit this phase relationship; the value is not a heat-transport travel time.

6. Step-size check: the finest 15/7.5 s pair differs by up to 0.2375 K across nodes and 0.04169 K at the response node, quantifying time-step sensitivity for this task.